

(C++) Computer Programming

Assignment - 2

1. Predict the output of the following program.

(a) **#include** <iostream>
using namespace std;

```
int main()
{
    int x = 8, y = 5;

    if (x > 5)
        if (y > 10)
            cout << "One";
        else
            cout << "Two";

    return 0;
}
```

(b) **#include** <iostream>
using namespace std;

```
int main()
{
    int x = 5, y = 10;

    if (x = y)
        cout << "Equal";
    else
        cout << "Not-Equal";

    return 0;
}
```

(c) **#include** <iostream>
using namespace std;

```
int main()
{
    int a = 10, b = 15, c = 20;

    if (a > b && a > c)
        cout << "A";
    else if (b > a && b > c)
        cout << "B";
    else
        cout << "C";

    return 0;
}
```

(d) **#include** <iostream>
using namespace std;

```
int main()
{
    int x = 4, y = -1, z = 0;

    cout << (x && y) << "-";
    cout << (x || z) << "-";
    cout << (z || y);

    return 0;
}
```

```
(e) #include <iostream>
    using namespace std;
```

```
int main()
{
    int a = 5, b = 10;

    if (a == b)
        if (a == 10)
            cout << "Hello";
        else
            cout << "Hi";

    return 0;
}
```

```
(f) #include <iostream>
    using namespace std;
```

```
int main()
{
    int a = 10, b = 20;

    if (a > 5)
        if (b < 10)
            cout << "A";
        else
            cout << "B";

    return 0;
}
```

2. If the ages of Ram, Shyam and Ajay are input through the keyboard, write a program to determine the youngest of the three.
3. If cost price and selling price of an item is input through the keyboard, write a program to determine whether the seller has made profit or incurred loss. Also determine how much profit he made or loss he incurred in percentage.
4. A library charges a fine for every book returned late. For first 5 days the fine is 50 paise, for 6-10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled. Write a program to accept the number of days the member is late to return the book and display the fine or the appropriate message.
5. Given the coordinates (x, y) of a center of a circle and it's radius, write a program which will determine whether a point lies inside the circle, on the circle or outside the circle.
6. Write a C++ program to input all three sides of a triangle and check whether the given sides can form a valid triangle or not. Display an appropriate message.
7. Write a C++ program to find all the roots of a quadratic equation. The program should determine whether the roots are real and distinct, real and equal, or complex roots, and display the appropriate results.

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